

Table 1

	Component name	Purpose	Location
1	Horizontal Arrangement	To arrange the child components horizontally	Palette--> Layout--> Horizontal Arrangement
2	Label	To display a label	Palette-->User Interface-->Label
3	Button	One button to serve as <i>play</i>	Palette-->User Interface-->Button
4	Button	One button to serve as <i>pause</i>	Palette-->User Interface-->Button
5	Button	One button to serve as <i>stop</i>	Palette-->User Interface-->Button
6	Player	To play the audio	Palette-->Media-->Player

So let's move on and add these functionalities using the block editor. The earlier article in this series discussed how to switch from the designer to the block editor. However, just to remind you, there is a button available right above the *Properties* pane to switch between the designer and block editor.

The block editor

I have already prepared the blocks for you. All you need to do is drag the relevant blocks from the left side palette and drop them on the viewer. Arrange the blocks the same way you see them in the image. I will explain what each block does and how it is called.

The blocks shown in Figure 3 are easy to understand. They just say 'play', 'pause' and 'stop' the player when the particular button is pressed. And that's it!

Now we are done with the block editor too. Next, we will move to downloading and installing the app on your phone to check how it is working.

Packaging and testing

To test the app, you need to get it on your phone. First, you have to download the application to your computer and then move it to your phone via Bluetooth or a USB cable. I'll tell you how to download the app.

On the top row, click on the *Build* button. It will give you the option of downloading the *apk* to your computer.

You can view the progress of the download and after it is successfully completed, the application will be placed in the *download* folder of your directory or in the location you have selected for your download.

Now you need to get this *apk* file to your mobile phone either via Bluetooth or via USB cable. Once you have placed the *apk* file on your SD card, you need to install it. Follow the on-screen instructions. You might get some notification or warning regarding installation from an untrusted source. Allow this from the *Settings* and after successful installation, you will see the icon of your application in the menu of your mobile.



Figure 1: The viewer

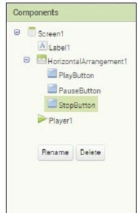


Figure 2: The components

Here you will see the default icon, which can be changed.

I hope your application is working exactly as per your design requirements. Now, depending upon your choice, you can customise various things like the image, sound and behaviour too.



Figure 3: Block editor

Debugging the application

We have just created the prototype of the application with very basic functionality. Now come the various cases that require serious attention so as not to annoy the user, and which your app should be able to sustain. Consider the cases below:

1. The app is coded to play a single audio file which the developer has uploaded to the server. Can it also access media files from local storage, like the SD card?
2. We have labelled the buttons with simple text. Can we, instead, put images over the buttons as symbols for *play*, *pause* and *stop*?
3. Can we add visuals or media information during the play?
4. Can we show how much of the music has been played in terms of duration (minutes or seconds)?

These are some of the scenarios that could possibly occur, and which the user would be pretty happy to see implemented. Think about how you can integrate these functions into the application. Do ask me if you fail to accomplish any of the above cases.

You have successfully built another useful Android app for yourself—a music player with three button controls. Happy inventing! 

By: Meghraj Singh Beniwal

The author is a B. Tech in electronics and communication, a freelance writer and an Android apps developer. He is currently working as a systems engineer at Infosys, Mysore. He can be contacted at meghraj/singh01@rediffmail.com